NS4168User ManualV0.9

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Change History

date	Version	author	Modify the description

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Figure Catalog

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1 Function Description

NS4168Is a supportI2SDigital audio signal input, output with anti-distortion function,2.5WMonoDClass A audio power amplifier.NS4168Especially suitable for environments that are sensitive to power consumption and produce interference. For example, Bluetooth speakers,WiFispeakers, tablet computers, etc. In these applications, system noise can corrupt the small analog signals sent to the amplifier.I2SThe standard is sent digitally to the audio amplifier, which significantly reduces the impact of noise sources on the transmitted audio; it also avoidsMCUThe main control chip has built-in audio decodingDACThe noise brought by the input coupling capacitor is eliminated, and a higher signal-to-noise ratio and lower distortion are finally obtained;CTRLThe pin detects a first-line pulse and selects the turning point of the internal input high-pass filter to match different speakers.layoutThere is no need to carefully consider the layout and routing of the audio amplifier, the periphery is simpler and debugging is more convenient.

NS4168Its unique anti-distortion function can effectively prevent output signal distortion caused by input signal overload and battery voltage drop, and can also effectively protect the speaker from being damaged during high-power output.

NS4168High efficiency, low noise modulation scheme is used, no externalLCThe closed-loop multi-level modulator design retains the high efficiency advantage of pure digital amplifiers while also having excellentPSRRand audio performance.DThe use of spread spectrum pulse density modulation provides lower electromagnetic radiation compared to similar architectures.NS4168exist5VWhen the working voltage is4ΩLoad provided2.5Woutput power.

NS4168It is a mono audio amplifier. The left and right channels are selected byCTRLPin level setting. Two chips can be selected for stereo products, which is very flexible.

NS4168Built-in over-current protection, over-heat protection and under-voltage protection functions can effectively protect the chip from being damaged under abnormal working conditions.eSOP8package, rated for operation over the temperature range of -40℃ to85℃.

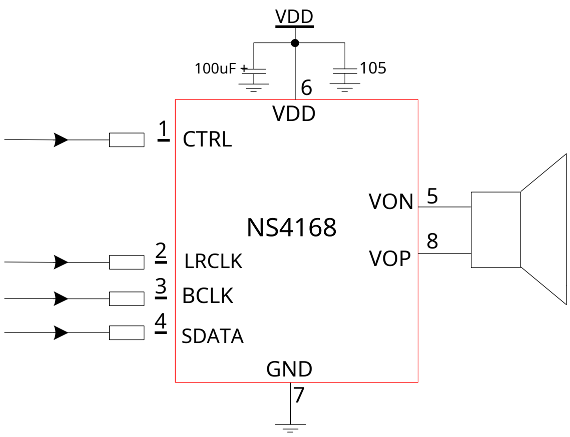
2 Key Features

- I2SSerial digital audio input interface
- Supports a wide range of sampling rates:8kHz~96kHz
- Automatic sampling rate detection, adaptive function
- Built-in digital high-pass filter, one-line pulse sets its turning point
- Left and right channels are optional, throughCTRLPin level setting
- Anti-aliasingNCNFunction,
- No filter requiredClass DAmplifier output power:
- 2.5W(VDD=5V, RL=4Ω) Operating voltage range:
- 3.0V~5.5V 0.2%THD（VDD=5V, RL=4Ω, Po=1W）
- 80%s efficiency(VDD=5V, RL=4Ω, Po=2.5W)
- Excellent "power-on, power-off" noise
- suppression
- Overcurrent protection, overheat protection, undervoltage protection
- eSOP8Encapsulation

3 Application Areas

- Bluetooth audio
- WiFiAudio
- Other portable speakers

4 Typical application circuit



picture1 NS4168Typical application circuit

5 Limit parameters

surface1 Chip maximum physical limit

parameter	Small value	Large value	unit	illustrate
voltage	2.8	5.5	V	
CTRL	- 0.3	VDD	V	
LRCLK	- 0.3	VDD	V	
BCLK	- 0.3	VDD	V	
SDATA	- 0.3	VDD	V	
ResistanceESDVoltage	4000		V	
Junction temperature	150		oC	
Recommended operating temperature	- 40	85	oC	
Recommended operating voltage	3	5.25		
Thermal resistance				
-JC(eSOP8)		20	oC/W	
-J A(eSOP8)		80	oC/W	
Soldering temperature -		220	oC	15Within seconds

Note: The chip's performance is not guaranteed outside of the limiting values or under any other conditions.

6 Electrical characteristics

Restrictions: (TA=25℃,VDD=5.0V)

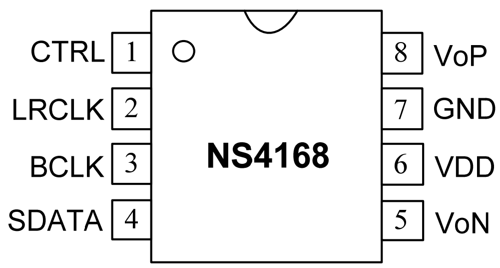
surface2 NS4168Electrical characteristics

symbol	parameter	Test Conditions	Small value	standard value	Large value	unit
V_{DD}	voltage		3		5. 25	V
I_{DD}	Power supply quiescent current	V_{DD}=5.0V, V_{IN}=0V,No load		13		mA
I_{SD}	Off leakage current	V_{CTRL}=0V		1		μA

V_{OS}	Output offset voltage			10	40	mV
PSRR	Power Supply Rejection Ratio	217Hz			- 80	dB
		20KHz			- 72	dB
CMRR	Common Mode Rejection Ratio			- 70		dB
f_{SW}	Modulation frequency	V_{DD}=3V to 5.25V		430		kHz
η	efficiency	P_o=2.5W,R_L=4Ω,		80		%
V_{CTRL}	CTRLThreshold	Right channel input	1.5		VDD	V
		Left channel input	0.9		1.15	
		ShutdownLow power shutdown			0.4	
T_{HI}	CTRLOne line pulse high Level time		1		12	us
T_{LO}	CTRLOne line pulse low Level time		1		12	us
T_{OFF}	CTRLOff time		100			us
t_{AT}	Attack time	V_{DD}=3.6V		10		ms
t_{RL}	Release time	V_{DD}=3.6V		1.1		s
P_O	Output Power	V_{DD}=3.6V		1.2		W
		V_{DD}=5V		2.5		W
THD+N	Total distortion + noise	f=1kHz R_L=4Ω, P_o=1.0W		0.2		%
SNR	Signal-to-Noise Ratio	RL=4Ω,Po=2.0W		85		dB
A_{MAX}	Large attenuation gain			- 10		dB

7 Chip pin description

7.1 NS4168Package pin assignment diagram



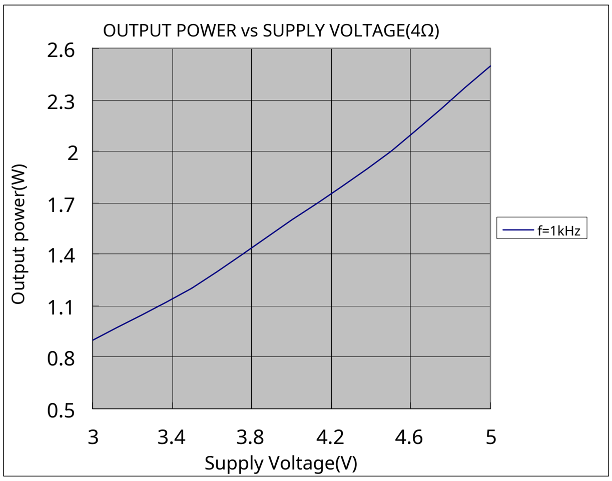
picture2 NS4168Package pin assignment diagram (top view)

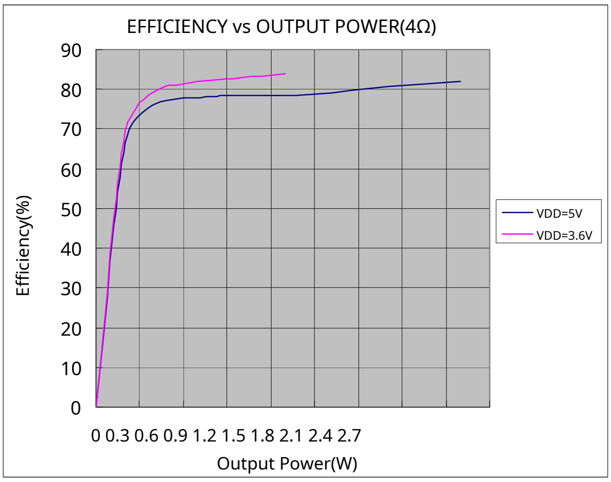
7.2 NS4168Pin Function Description

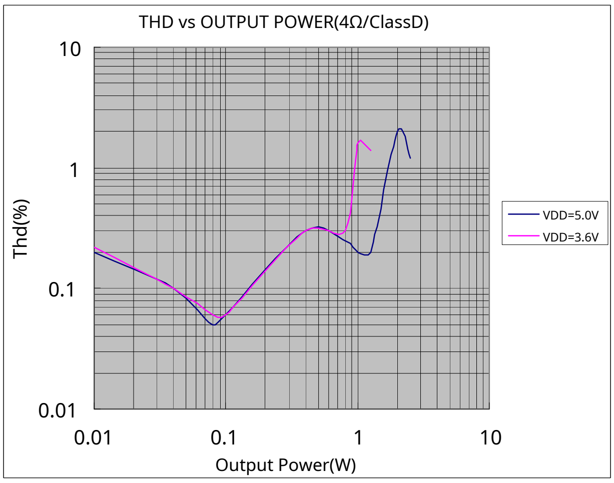
surface3 NS4168Pin Description

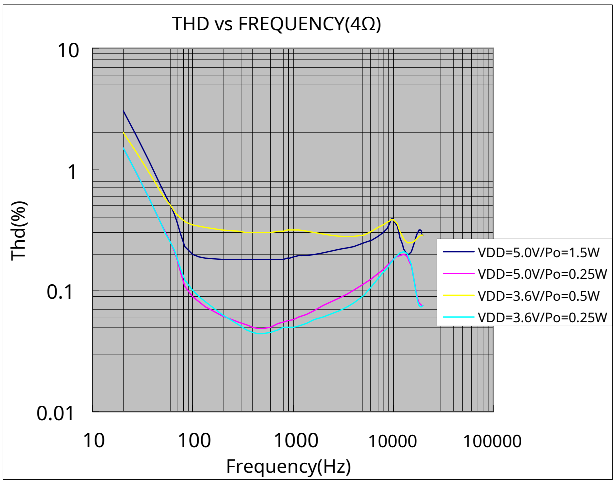
symbol	Pin Number	describe
CTRL	1	Control pin (see below for details)
LRCLK	2	I2SLeft and right channel frame clock
BCLK	3	I2SBit Clock
SDATA	4	I2SSerial Data
V N	5	Output negative terminal
VDD	6	power input
GND	7	Power Ground
VoP	8	Output positive terminal

8 NS4168Typical Reference Characteristics





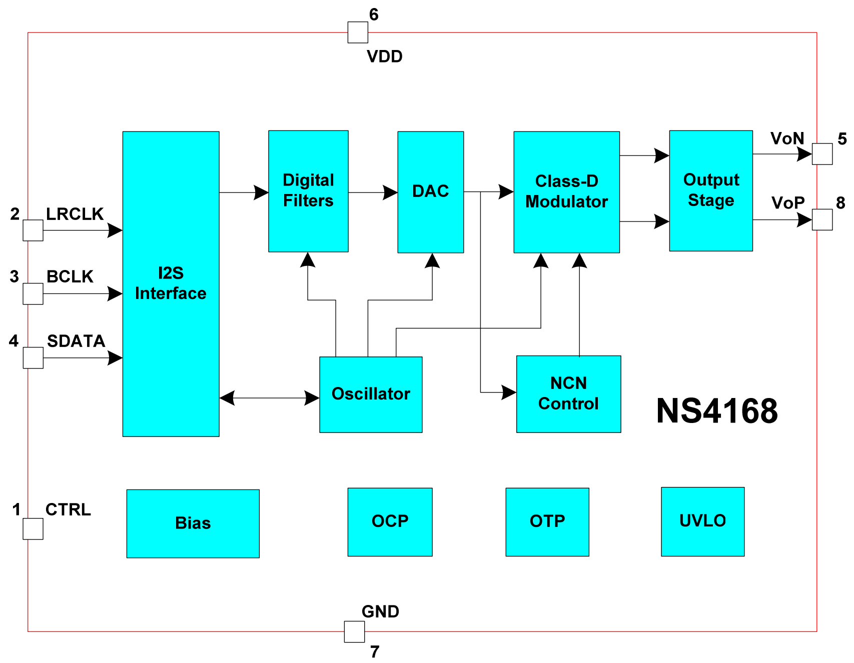




9 NS4168Application Notes

9.1Description of the basic structure of the chip

NS4168Is a supportI2SDigital audio signal input, output with anti-distortion function,2.5WMonoDClass A audio power amplifier, bridge output. Its principle block diagram is as follows:

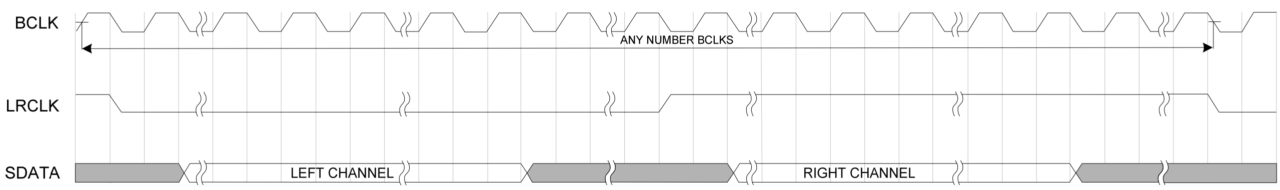


picture3 NS4168Block Diagram

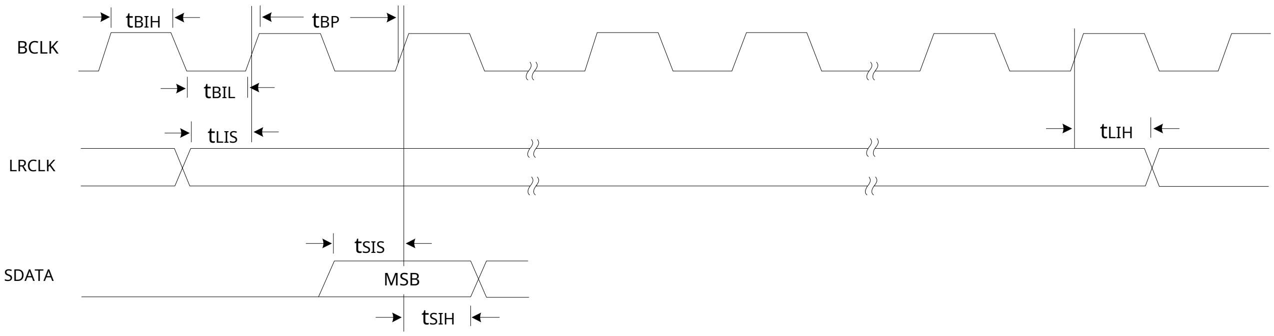
9.2 I2SDigital input serial audio interface

9.2.1 I2SDigital Audio Formats

NS4168supportI2SDigital audio signal input. StandardI2Shave3Main signals: serial clockBCLK, frame clockLRCLK, Serial DataSDATASerial ClockBCLKAlso called bit clock, which corresponds to each bit of digital audio data.LRCLKData used to switch the left and right channels.LRCLKfor"1"Indicates that the data of the right channel is being transmitted, which is "0”It means that the data of the left channel is being transmitted.LRCLKThe frequency of the serial data is equal to the sampling frequency.SDATAIt is audio data represented by two's complement.I2SThe digital audio format is as follows:



picture4 I2SDigital Audio Formats



picture5 I2SDigital Audio Timing Diagram

surface4Timing parameter table

parameter	Small value	unit	describe
t_{BIL}	40	ns	BCLKLow level pulse width
t_{BIH}	40	ns	BCLKHigh level pulse width
t_{LIS}	10	ns	LRCLKorSDATAEdge toBCLKRising edge setup time
t_{LIH}	10	ns	BCLKRising edge toLRCLKorSDATAEdge hold time
t_{SIS}	10	ns	SDATAarriveBCLKRising edge setup time
t_{SIH}	10	ns	BCLKRising edge toSDATAHold time

9.2.2 Left and right channel settings

NS4168For a mono amplifier, the standardI2SThe protocol can transmit left and right channel signals.NS4168passpin1（CTRL） pin level setting to select left or right channel signal. See the relevant section below for details.

9.3Input channel selection and input high-pass filter setting

NS4168Input channel selection and input high-pass filter turning point setting are all done throughCTRLPin implementation.

9.3.1 Input channel selection

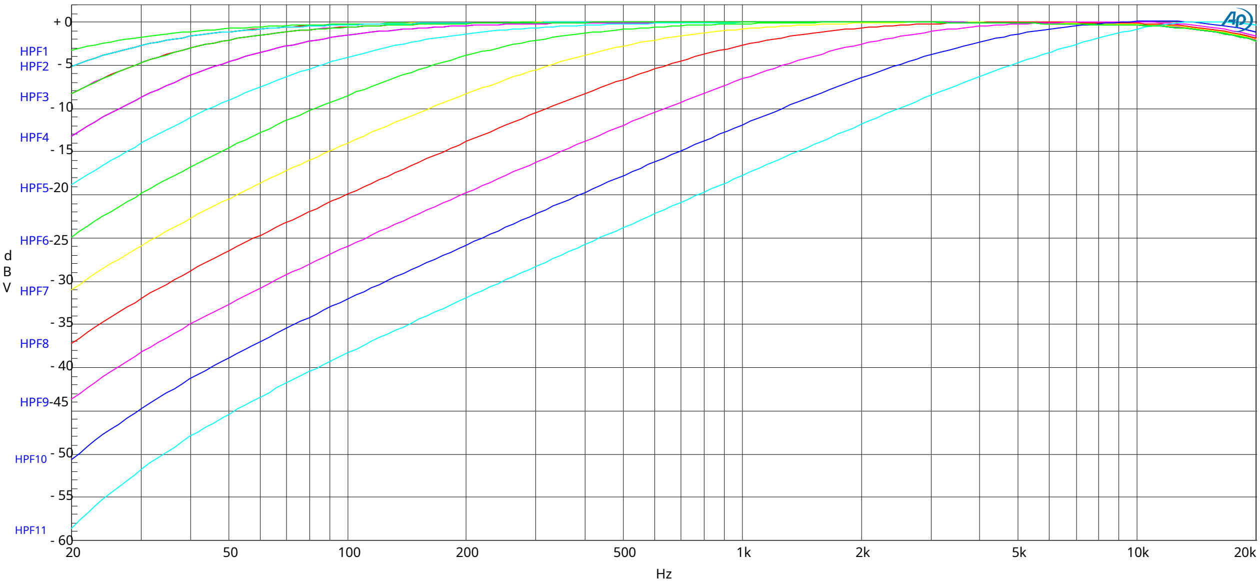
NS4168Input channel selection viaCTRLThe level setting of the pin.CTRLThe pin voltage is0.9V~1.15V, select the left channel; CTRLThe pin voltage is1.5 V ~ VDD When the right channel is selected, the right channel is selected. As shown in the following table:

surface5Channel selection settings

CTRL pin voltage	Channel
0.9V~1.15V	Left Channel
1.5 V ~ VDD	Right Channel

9.3.2 Input High Pass Filter Settings

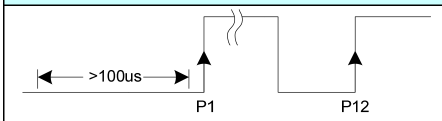
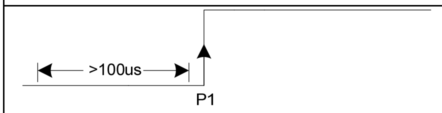
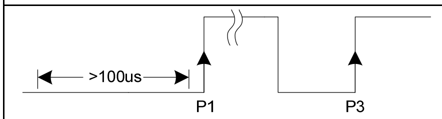
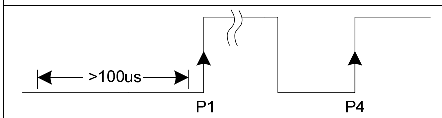
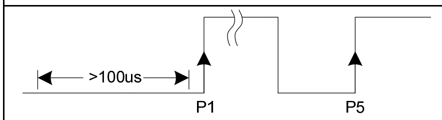
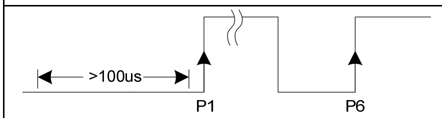
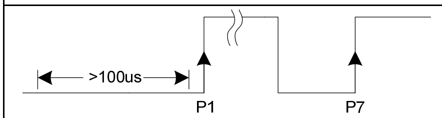
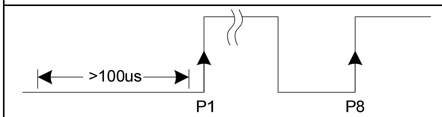
NS4168The input has a high-pass filter, and the filter's corner frequency passesCTRLPin one line pulse setting, a total of11 There are two options:HPF1~HPF11.As shown below(VDD=5V):

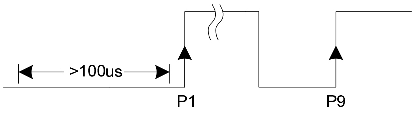
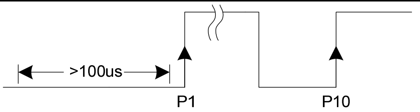
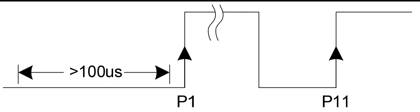


picture6 NS4168Input high pass filter frequency response curve

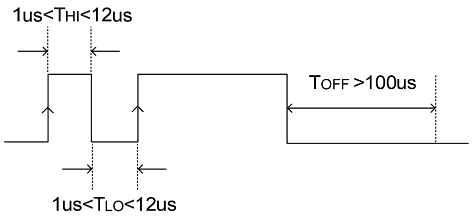
The corresponding number of pulses and waveforms for one line are as follows:

surface6Correspondence between a line pulse waveform and a high-pass filter

CTRLPin one line pulse waveform	Number of rising edges of one-line pulse	Corresponding high pass filter	filter-3dBTurning Point (Typical value)
	12	HPF1	20Hz
	1	HPF2	28Hz
	3	HPF3	40Hz
	4	HPF4	65Hz
	5	HPF5	120Hz
	6	HPF6	240Hz
	7	HPF7	458Hz
	8	HPF8	910Hz

	9	HPF9	1.82kHz
	10	HPF10	3.5kHz
	11	HPF11	6.6kHz

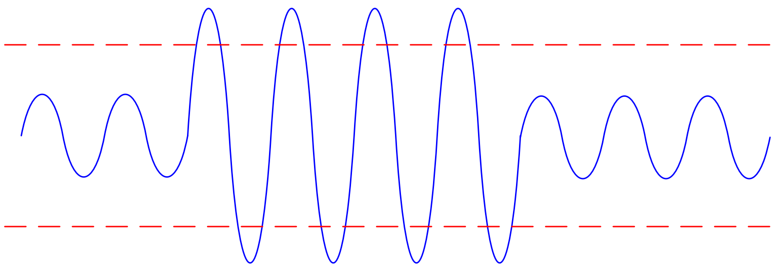
Add in toCTRLThe high level width of the pin's pulse line (T_{HI})Require1us<T_{HI}<12us. Low level width (T_{LO})Require 1us<T_{LO} <12us.EnterSHUTDOWNMode low level hold time (T_{OFF})RequireT_{OFF}>100usThe timing diagram is as follows:



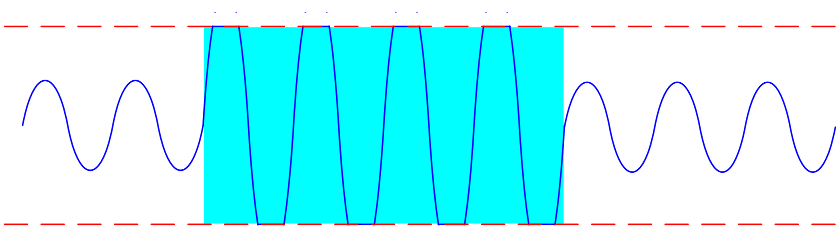
picture7One-line pulse timing diagram

9.4Anti-aliasingNCN)Function

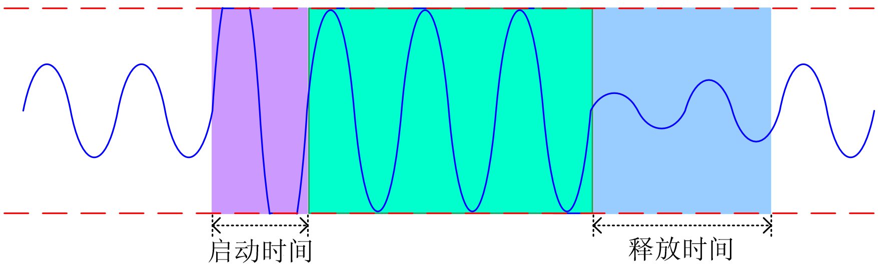
NS4168It has an anti-distortion function. The anti-distortion function can effectively prevent the output signal from being distorted due to input signal overload and battery voltage drop. It can effectively protect the speaker from damage when high power output is applied. The principle is that the amplifier automatically detects output clipping distortion and automatically adjusts the amplifier. The gain of the amplifier is increased to achieve the anti-distortion effect. As shown in the following figure:



picture8Audio output signal assuming no power supply voltage limitation



picture9Audio output signal in normal working mode



picture10Audio output signal in anti-distortion working mode

9.5Power filter capacitor selection

In amplifier applications, power supply bypass design is very important, especially for the noise performance and power supply voltage suppression performance of the application solution. The design requires the filter capacitor to be as close to the chip power pin as possible. The typical capacitor is100uFThe capacitor is connected1uFof ceramic capacitors.

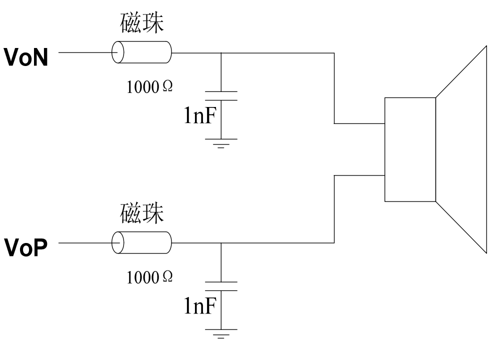
9.6protect the circuit

When the chip output pin is short-circuited to the power supply or ground, or the output is short-circuited, the overcurrent protection circuit will shut down the chip to prevent the chip from being damaged.NS4168Automatically resume work. When the chip temperature is too high, the chip will also be shut down. After the temperature drops,NS4168Continue to work normally. When the power supply voltage is too low, the chip will also be shut down, and the chip will start again after the power supply voltage is restored.

9.7 layoutsuggestion

NS4168 is a Class D amplifier, and EMI interference should be considered during layout. The following aspects can be used to minimize EMI interference:

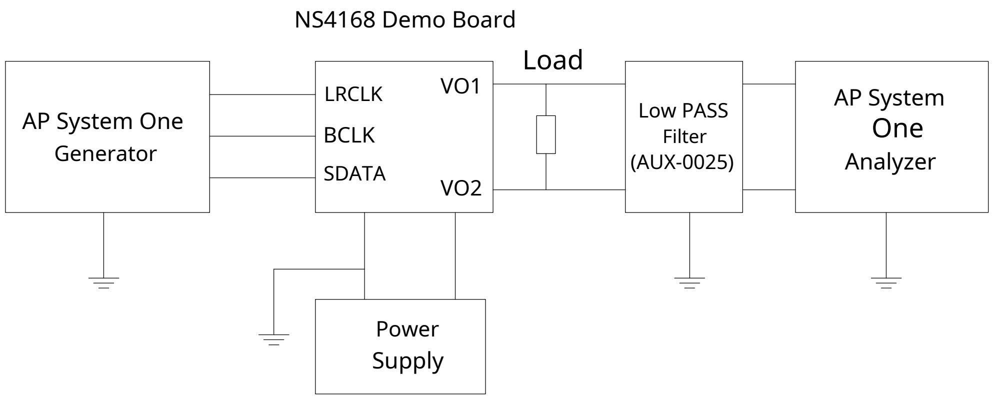
1. The wiring from the amplifier output to the speaker should be as short and wide as possible, and the output wiring and wiring should be as far away from sensitive signal lines and circuits as possible.
2. The decoupling capacitor of the power pin of the power amplifier should be as close to the chip pin as possible. The power line and ground line should be connected in star shape.
3. When the application environment is relatively harsh due to space limitations, adding magnetic beads and capacitors at the output end can effectively suppressEMIInterference. When using, the magnetic beads and capacitors should be as close to the chip pins as possible. The following areNS4168The application design reference circuit after adding magnetic beads and capacitors to the output end:



picture11Application diagram of adding magnetic beads to the output end

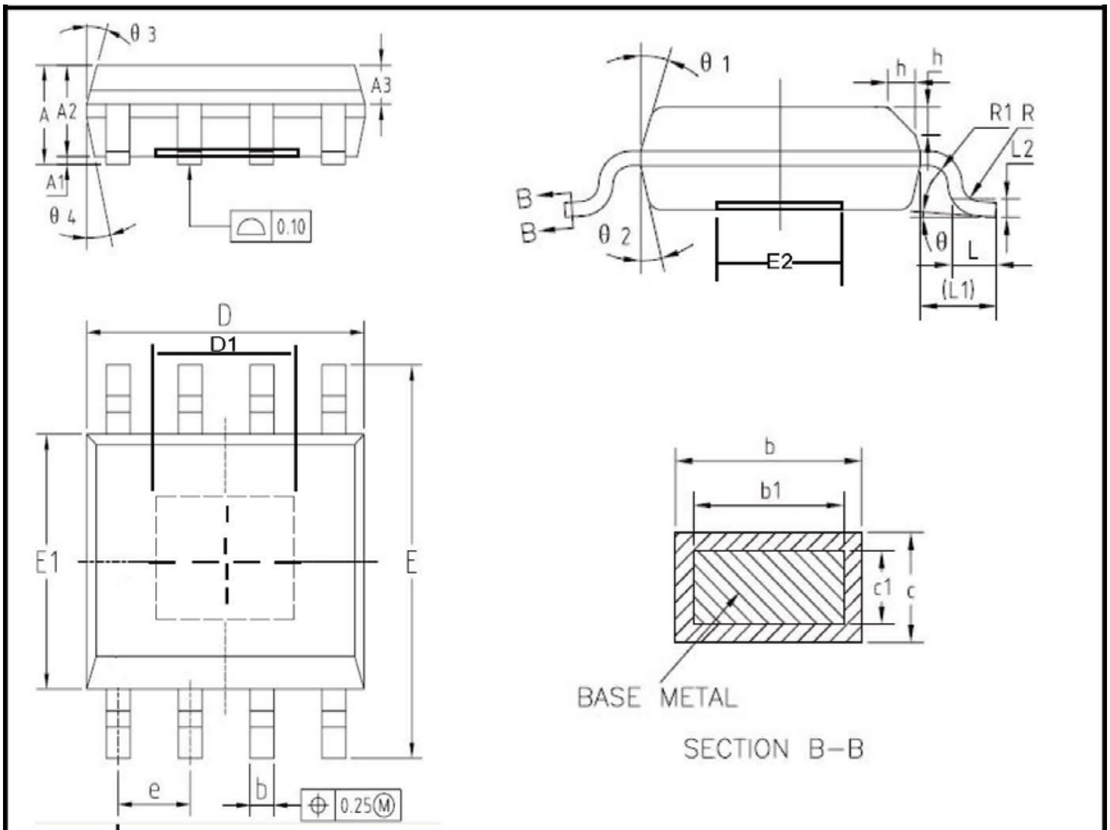
9.8Test Circuit

NS4168The test circuit is as shown below.DWhen the amplifier is in class mode, the low-pass filter (Low PASS Filter)It is necessary. You can use two33uHThe inductor is connected in series at both ends of the load resistor to be equivalent to a speaker. If only a pure resistor is used to replace the speaker load, the measured results will be worse than when the speaker is used as a load, including power, efficiency, distortion and other indicators.



picture12 NS4168Test Circuit

10 Chip packaging



	MIN	NOM	MAX		MIN	NOM	MAX
A	1.450	1.550	1.650	e		1.270	
A1	0.100	0.150	0.200	L	0.450	0.600	0.750
A2	1.250	1.400	1.650	L1		1.040	
A3	0.550	0.600	0.650	L2		0.250	
b	0.380	–	0.510	R	0.070	–	–
b1	0.370	0.420	0.470	R1	0.070	–	–
c	0.170	–	0.250	h	0.300	0.400	0.500
c1	0.170	0.200	0.230	θ	0.000	–	6.000
D	4.850	4.900	4.950	θ 1	15.000	17.000	19.000
E	5.850	6.000	6.150	θ 2	11.000	13.000	15.000
E1	3.850	3.900	3.950	θ 3	15.000	17.000	19.000
E2	1.9	2.0	2.2	θ 4	11.000	13.000	15.000
D1	1.9	2.0	2.2				

picture13 eSOP-8Package Dimensions